Mezzanine funds hold portfolios of loans in private companies. The debt held by these funds tends to be low on the totem pole; the term "mezzanine" implies that it is ranked below senior debt. Some mezzanine funds work hand in hand with buyout or real estate funds. This is a place to park some of the debt issued by private companies.

Infrastructure funds concentrate in capital infrastructure investments including airports, railways highways, and utilities. Typically these are leveraged.

A *distressed* fund concentrates on sick companies that are in or near bankruptcy. GPs of these funds specialize in *turnaround management,* which tries to turn sick companies into healthy ones.

Funds-of-funds are like their counterparts in hedge-fund land: they hold investments in different PE funds. Some funds-of-funds are listed.[8]

PE *secondary funds* acquire preexisting investments in PE funds from LPs. They allow existing LPs an opportunity to exit. (I discuss the secondary PE market in chapter 13.) There was a burst of secondary fund raisings during the financial crisis, as some savvy investors took advantage of distressed LPs.

LPs often invest directly in individual companies, alongside the PE funds, through *co-investments*.

2.4. PRIVATE EQUITY COMMITMENTS ARE PRO-CYCLICAL

Institutional asset owners tend to invest in PE pro-cyclically, injecting money into PE funds right at the peak of the business.[9] (These institutions have the same pro-cyclical tendencies as individuals investing in mutual funds; see chapter 16). Money piles in when current valuations are high and past distributions are large—but future expected returns are low. Paul Gompers and Joshua Lerner (2000), two prominent professors specializing in PE, call this "money chasing deals": asset owners tend to put money in PE right when it is most expensive.

Figure 18.2 shows PE fundraising over the 2000s decade as compiled by Pitchbook, a PE database. Panel A shows capital raised at around $320 billion per year peaking in 2007 and 2008, right before the financial crisis and the ensuing economic slowdown. Then in 2009 and 2010, right at the time when prices were low and expected returns were high, PE equity fundraising fell off a cliff. Panel B shows a similar picture in terms of the number of funds closed. (The correlation between the numbers in Panels A and B is approximately 90%).

[8] The organizational forms of listed versus unlisted private equity investment vehicles are different. Listed PE funds typically have indefinite lives and stricter reporting requirements. Jensen (2007) argues that reputational concerns give unlisted PE funds additional incentives to perform and therefore such funds should have higher returns than listed ones.

[9] See evidence documented by Gompers and Lerner (2000) and Kaplan and Lerner (2010) for VC and Kaplan and Stromberg (2009) and Axelson et al. (2012) for buyouts.